1D Finite Element Method (FEM)

Prereq: IB HL 3-4 Applications / Analysis, Intuition with heat equations, Approx time for presentation 30min

May 2026

Motivation

The Problem: How to solve a PDE (Physics Differential Equation)?

The Reality: Computers can only approximate calculus with finite differences.

Sample PDE:

$$-k\frac{d^2T}{dx^2} = Q$$

- k = conductivity rate
- Q = external heat generation
- T = temperature (what we want)
- x = location on the bar

(Anyone who said we can integrate twice to solve analytically is banned.)

Approach 1: Noob ML

- Guess a solution, approximate the PDE loss, then update the guess via gradient descent.

Cons:

- Approximating PDE loss is tacky for higher-order derivatives.
- The network can get stuck in local minima.
- It takes forever to train.

Approach 2: Finite Element Method (FEM) (Sigma Method)

Step 1: Find the Weak Form

"Strong form" refers to the actual PDE.

Weak form tries to make the **average residual** (mistakes) zero:

$$\int_0^L R(x)dx = 0 \quad \text{where} \quad R(x) = -k\frac{d^2T}{dx^2} - Q$$

The Trap: One place may have a $+5$ mistake, another a -5 mistake. They average to zero :DDDDD

The Fix: Multiply both sides by an arbitrary *test function* $v(x)$:

$$\int_0^L R(x)v(x)dx = 0$$

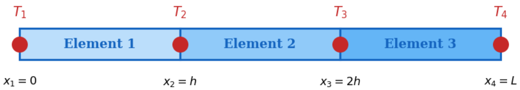
If this holds true for *enough* variations of $v(x)$, then $R(x)$ must be exactly 0 everywhere!
Proof: Intuition. You can also use Dirac Deltas.

Step 2: Discretize the 1D Bar

Place nodes where we want to solve for temperature. Let's use equispaced nodes on a bar from $[0, L]$.

- **Lengths (Known):** $0, h_1, h_2, h_3 \ldots h_{n-1}, L$
- **Temperatures (Unknown):** $T_0, T_1, T_2, T_3 \ldots T_n$

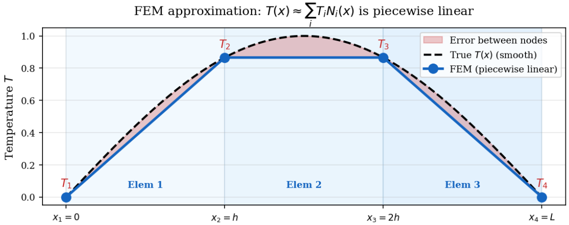
Discretization of a 1D Bar into Linear Elements



Cont.

Approximate $T(x)$ with linear elements(shape functions $N_i(x)$):

- From 0 to h: $T(x) = T_0 N_0(x) + T_1 N_1(x)$
- From h to $2h$: $T(x) = T_1 N_1(x) + T_2 N_2(x)$
- From $2h$ to L: $T(x) = T_2 N_2(x) + T_3 N_3(x)$

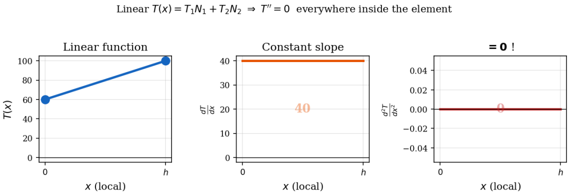


Step 3: Solve for One Little Slice

Let's zoom in on the first slice: 0 to h_1.

$$-k \int_0^{h_1} \frac{d^2 T}{dx^2} v(x) dx = \int_0^{h_1} Q v(x) dx$$

If we substitute $T(x) = T_0 N_0(x) + T_1 N_1(x)$, we are cooked, because the second derivative $\frac{d^2 T}{dx^2}$ instantly becomes 0.



Solution: Integration by Parts.

Step 3.5: The Actual Integration by Parts

To use linear elements, we must reduce the order of $\frac{d^2T}{dx^2}$.

Recall the IBP formula: $\int u\,dw = [uw] - \int w\,du$.

Let's define our parts for the LHS:

- Let $u = v(x) \implies du = \frac{dv}{dx}\,dx$
- Let $dw = -k\frac{d^2T}{dx^2}\,dx \implies w = -k\frac{dT}{dx}$

We reduce order of $\frac{d^2T}{dx^2}$, while differentiate $v(x)$, which we can control to be something easy to differentiate. Math got scammed by IBP.

Cont.

Substituting back into the integral for the first slice $[0, h_1]$:

$$\int_0^{h_1} \left(-k\frac{d^2T}{dx^2}\right) v(x)\, dx = \left[-kv(x)\frac{dT}{dx}\right]_0^{h_1} - \int_0^{h_1} \left(-k\frac{dT}{dx}\right) \frac{dv}{dx}\, dx$$

$$\cdots = \left[-kv(x)\frac{dT}{dx}\right]_0^{h_1} + k\int_0^{h_1} \frac{dT}{dx}\frac{dv}{dx}\, dx$$

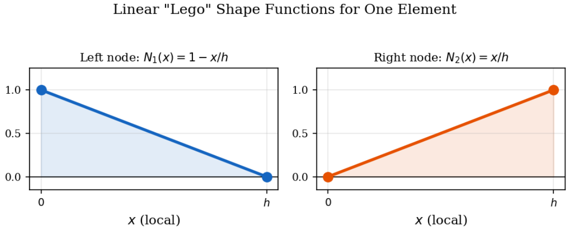
The Result: We successfully shared the derivative! T no longer needs a second derivative to be solved.

Step 4a: Meet the "Lego" Blocks

To actually do the integration, we need to define our linear shape functions for a single slice (element) of length h.

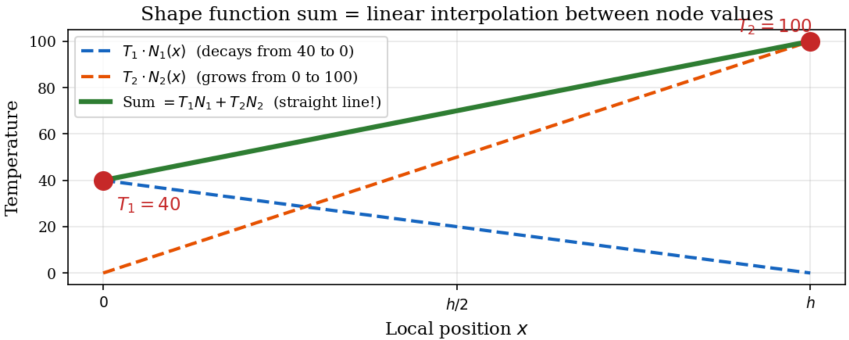
Let the local coordinate x go from 0 (left node) to h (right node).

- **Left Node Function (N_1):** Starts at 1, goes to 0.
  $N_1(x) = 1 - \frac{x}{h} \implies \frac{dN_1}{dx} = -\frac{1}{h}$
- **Right Node Function (N_2):** Starts at 0, goes to 1.
  $N_2(x) = \frac{x}{h} \implies \frac{dN_2}{dx} = \frac{1}{h}$



Cont.

Note that $T(x) \approx T_1 N_1(x) + T_2 N_2(x)$ is just linear interpolation between T_1 and T_2



Step 4b: Substituting our "Legos"

Recall our weak form: $k\int_0^h \frac{dT}{dx}\frac{dv}{dx}dx = \int_0^h Q \cdot v(x)dx$

First, we substitute our actual temperature guess for this slice:

$$T(x) = T_1 N_1(x) + T_2 N_2(x) \implies \frac{dT}{dx} = T_1\frac{dN_1}{dx} + T_2\frac{dN_2}{dx}$$

Now let's set $v(x)$ to some convenient values to generate a system of equations.

- **Test 1:** Let $v(x) = N_1(x) \implies \frac{dv}{dx} = \frac{dN_1}{dx}$
- **Test 2:** Let $v(x) = N_2(x) \implies \frac{dv}{dx} = \frac{dN_2}{dx}$

Step 4c: Expanding the Integral

Equation 1 (Testing with N_1):
Substitute our T expansion and $v = N_1$ into the weak form:

$$k \int_0^h \left(T_1 \frac{dN_1}{dx} + T_2 \frac{dN_2}{dx} \right) \frac{dN_1}{dx} dx = \int_0^h Q \cdot N_1 dx$$

Distribute the integral to isolate our unknown temperatures (T_1, T_2):

$$T_1 \underbrace{\left(k \int_0^h \frac{dN_1}{dx} \frac{dN_1}{dx} dx \right)}_{K_{11}} + T_2 \underbrace{\left(k \int_0^h \frac{dN_2}{dx} \frac{dN_1}{dx} dx \right)}_{K_{12}} = \underbrace{\int_0^h Q \cdot N_1 dx}_{F_1}$$

Equation 2 (Testing with N_2):
Doing the exact same thing, but multiplying by dN_2/dx gives us:

$$T_1 \underbrace{\left(k \int_0^h \frac{dN_1}{dx}\frac{dN_2}{dx}dx\right)}_{K_{21}} + T_2 \underbrace{\left(k \int_0^h \frac{dN_2}{dx}\frac{dN_2}{dx}dx\right)}_{K_{22}} = \underbrace{\int_0^h Q \cdot N_2 dx}_{F_2}$$

Step 4d: Collapsing to Matrix Form

Therefore

$$K_{11}T_1 + K_{12}T_2 = F_1$$

$$K_{21}T_1 + K_{22}T_2 = F_2$$

Which can be rewritten as

$$\begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

When we actually plug in our constant derivatives ($\frac{dN_1}{dx} = -\frac{1}{h}$ and $\frac{dN_2}{dx} = \frac{1}{h}$) and evaluate those integrals, we get the final local system:

$$\frac{k}{h} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \end{bmatrix} = \frac{Qh}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Step 5a: Global Assembly (The Setup)

Now we have the math for one slice. Let's look at a bar divided into **two slices** (Elements A and B), meaning we have **three nodes** total.

- **Element A** connects Node 1 and Node 2.
- **Element B** connects Node 2 and Node 3.

How do we combine their local 2×2 matrices into a massive 3×3 global matrix for the whole bar?

Rule: You just add them together wherever they share a node.

Step 5b: Justifying the Assembly (Nodal Equilibrium)

Let's look at the underlying equations, mapping each element's terms directly to its global node numbers (1, 2, and 3).

Element 1 (Connects Nodes 1 and 2):

$$K_{11}T_1 + K_{12}T_2 = F_1$$

$$K_{21}T_1 + K_{22}T_2 = F_2$$

Element 2 (Connects Nodes 2 and 3):

$$K_{22}T_2 + K_{23}T_3 = F_2$$

$$K_{32}T_2 + K_{33}T_3 = F_3$$

Cont.

At **Node 2**, the total heat flux must balance. Adding the internal fluxes from both elements gives the combined equation for the middle node:

$$(K_{21}T_1 + K_{22}T_2) + (K_{22}T_2 + K_{23}T_3) = F_{2,total}$$

Grouping the T_2 terms gives us the overlapping middle row of our matrix!

$$K_{21}T_1 + (\mathbf{K_{22}} + \mathbf{K_{22}})T_2 + K_{23}T_3 = F_{2,total}$$

Step 5c: Expanding to Global Coordinates

Because we named our terms using their global node numbers, we can simply drop each local matrix into a 3×3 global grid padded with zeros.

Element 1 (Connects Nodes 1 & 2, missing 3):

$$[K_1] = \frac{k}{h} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \{F_1\} = \frac{Qh}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Element 2 (Connects Nodes 2 & 3, missing 1):

$$[K_2] = \frac{k}{h} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}, \quad \{F_2\} = \frac{Qh}{2} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Cont.

Now, the global assembly is just direct matrix addition:

$$[K_{global}] = [K_1] + [K_2]$$

$$\{F_{global}\} = \{F_1\} + \{F_2\}$$

Step 5d: Global Matrix

$$[K] = \frac{k}{h}\begin{bmatrix} 1 & -1 & 0 \\ -1 & (\mathbf{1}+\mathbf{1}) & -1 \\ 0 & -1 & 1 \end{bmatrix} = \frac{k}{h}\begin{bmatrix} 1 & -1 & 0 \\ -1 & \mathbf{2} & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\{F\} = \frac{Qh}{2}\begin{bmatrix} 1 \\ \mathbf{1}+\mathbf{1} \\ 1 \end{bmatrix} = \frac{Qh}{2}\begin{bmatrix} 1 \\ \mathbf{2} \\ 1 \end{bmatrix}$$

Now we can solve $[K]\{T\} = \{F\}$ using $\{T\} = [K]^{-1}\{F\}$.

Step 6: The Boundary Conditions

Remember back in Step 3.5 when we did Integration by Parts? We completely ignored this bracketed term:

$$\left[-kv(x)\frac{dT}{dx}\right]_0^L$$

We can't ignore it anymore (just like your problems). This term represents the **Boundary Conditions (BCs)** at the absolute ends of our bar. There are two main types of physical boundaries we need to deal with:

- **Dirichlet:** We know the exact temperature at the boundary (e.g., holding an ice cube at the end, so $T = 0$). For the integrating twice charlatans, this is the $+C$.
- **Neumann:** We know the heat flux/flow at the boundary (e.g., the end is wrapped in thick insulation, so heat flow is 0). This is the $+Dx$.

Applying Neumann BCs (Free Math)

Let's say the right end of the bar $(x = L)$ is **perfectly insulated**.

What does insulation mean physically? It means heat cannot flow in or out. Mathematically, the slope of the temperature curve must be zero:

$$\frac{dT}{dx} = 0$$

Plug that into our scary boundary bracket:

$$[-kv(L)(0)] = 0$$

The Result: The entire term vanishes! We literally do nothing to our global matrix. The weak form naturally handles insulated boundaries for free.

Applying Dirichlet BCs (The Matrix Hack)

What if the left end of the bar ($x = 0$) is held at exactly 100°C? We already have Node 1 representing $x = 0$, so we need to force $T_1 = 100$.

We simply **hack the matrix**. We go to row 1 of our global system and overwrite it:

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} \mathbf{100} \\ F_2 \\ F_3 \end{bmatrix}$$

Look at the top row: $1 \cdot T_1 + 0 \cdot T_2 + 0 \cdot T_3 = 100$.

We just hardcoded reality into the linear algebra :D

Enough Yap Let's Solve

The Problem: A 3-node bar. Uniform internal heat generation. Left end fixed at 100°C. Right end insulated.

- Let our stiffness multiplier $\frac{k}{h} = 1$
- Let our heat load vector for each element $\frac{Qh}{2} = 10$

Our Assembled Global System (before BCs):

$$1 \cdot \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 + 10 \\ 10 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \\ 10 \end{bmatrix}$$

Let's Actually Solve One: The Execution

1. **Apply BCs:** Node 1 is 100°C (Matrix Hack). Node 3 is insulated (Do nothing).

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} \mathbf{100} \\ 20 \\ 10 \end{bmatrix}$$

2. **Use peasant algebra (no matrix inversion?)**
   - **Row 1:** $T_1 = 100$
   - **Row 3:** $-T_2 + T_3 = 10 \implies T_3 = T_2 + 10$
   - **Row 2:** $-T_1 + 2T_2 - T_3 = 20$

Substitute T_1 and T_3 into Row 2:

$$-(100) + 2T_2 - (T_2 + 10) = 20 \implies T_2 - 110 = 20 \implies \mathbf{T_2 = 130}$$

$$T_3 = 130 + 10 \implies \mathbf{T_3 = 140}$$

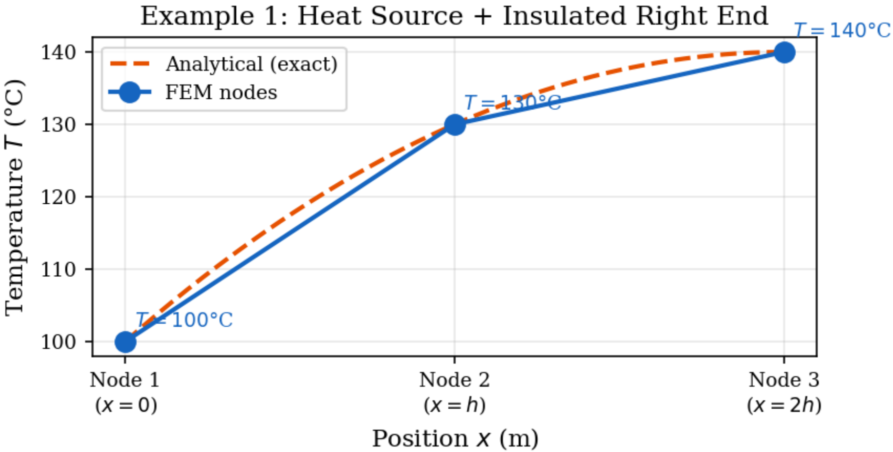
Did it work?

Hence $T_1 = 100°\text{C}, \quad T_2 = 130°\text{C}, \quad T_3 = 140°\text{C}$

Does this make physical sense?

- Heat is being generated everywhere inside the bar (Q).
- The right side (T_3) is insulated, so heat is trapped and starts piling up.
- The left side (T_1) is fixed at $100°\text{C}$, acting as a refrigerator keeping that side cool.

Because heat flows from hot to cold, the heat must flow from right to left. Therefore, the right side **must** be the hottest part of the bar. $140 > 130 > 100$.


Example 1: Heat Source + Insulated Right End
Analytical (exact)
FEM nodes
$T = 140°C$
$T = 130°C$
$T = 100°C$
140
130
120
110
100
Temperature T (°C)
Node 1
$(x = 0)$
Node 2
$(x = h)$
Node 3
$(x = 2h)$
Position x (m)

Example 2: The Asymmetric Radiator

Let's break the symmetry. We have a 3-node bar ($k = 2$), but the slices are different lengths and have different heat sources.

- **Element 1 (Nodes 1 to 2):** Long and actively heating up.
  $h_1 = 2\text{m}, \quad Q_1 = 4 \text{ W/m}^3 \implies \frac{k}{h_1} = 1, \quad F_{elem1} = \frac{Q_1 h_1}{2} = 4$
- **Element 2 (Nodes 2 to 3):** Short and inactive.
  $h_2 = 1\text{m}, \quad Q_2 = 0 \text{ W/m}^3 \implies \frac{k}{h_2} = 2, \quad F_{elem2} = 0$

Global Assembly: Node 2 gets stiffness from both, but heat only from Element 1.

$$[K] = \begin{bmatrix} 1 & -1 & 0 \\ -1 & (1+2) & -2 \\ 0 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 2 \end{bmatrix}, \quad \{F\} = \begin{bmatrix} 4 \\ 4+0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix}$$

Example 2: Solve and Intuition

The Boundary Conditions: Left end is cold ($T_1 = 10$). Right end is warm ($T_3 = 50$).

Hack the matrix for Dirichlet boundaries:

$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ -1 & 3 & -2 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} \mathbf{10} \\ 4 \\ \mathbf{50} \end{bmatrix}$$

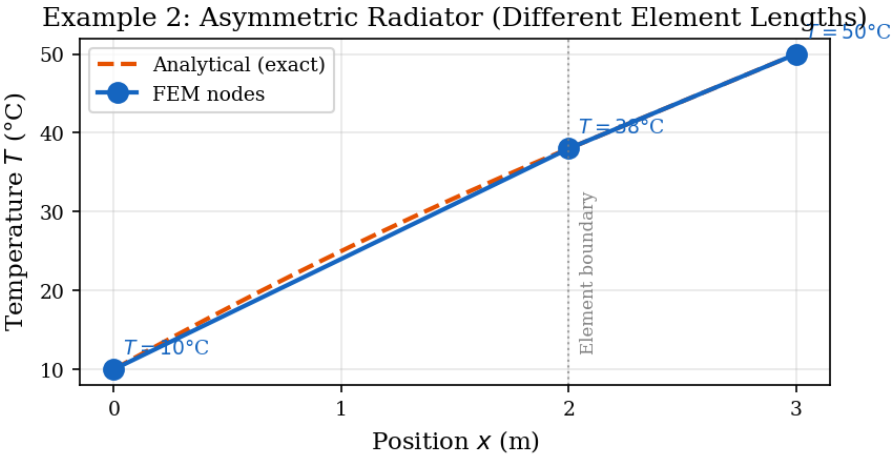
Solve Row 2:

$$-T_1 + 3T_2 - 2T_3 = 4 \implies -(10) + 3T_2 - 2(50) = 4$$

$$3T_2 - 110 = 4 \implies 3T_2 = 114 \implies \mathbf{T_2 = 38}$$

Physical Intuition Check: Without internal heat, the temperature would just linearly interpolate between 10 and 50. Since Node 2 is 2/3 of the way down the bar, it "should" naturally be 36.6. But because Element 1 is actively generating heat ($Q_1 = 4$), it pushes Node 2 slightly hotter, up to 38.

Cont.


Example 2: Asymmetric Radiator (Different Element Lengths)
Analytical (exact)
FEM nodes
T = 10°C
T = 38°C
T = 50°C
Element boundary
Temperature T (°C)
Position x (m)
50
40
30
20
10
0
1
2
3

What if Q is negative? Let's blow on the bar to cool it down.

- **Parameters:** $k = 1, \quad h_1 = 1, \quad h_2 = 1$
- **Heat Sink:** $Q = -10$ W/m^3 everywhere.

Global Assembly:

$$\frac{k}{h} = 1, \quad \frac{Qh}{2} = -5 \implies [K] = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}, \quad \{F\} = \begin{bmatrix} -5 \\ -10 \\ -5 \end{bmatrix}$$

Example 3: Solve and Intuition

The Boundary Conditions: Both ends are clamped to boiling water ($T_1 = 100$, $T_3 = 100$).

Hack the matrix:

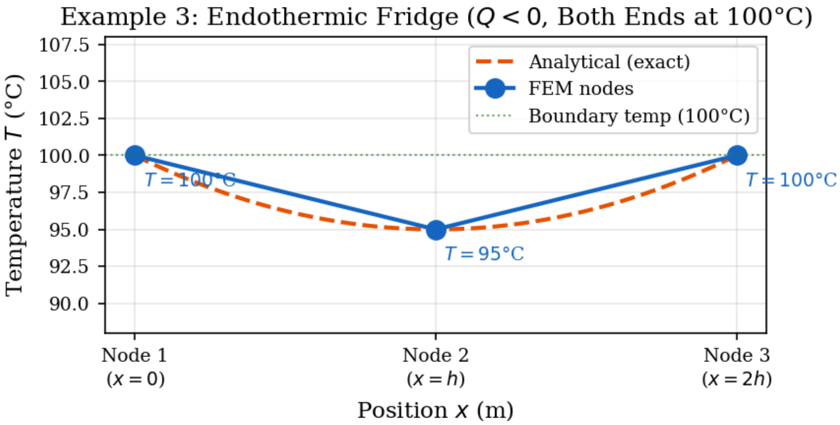
$$\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ -1 & 2 & -1 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} \mathbf{100} \\ -10 \\ \mathbf{100} \end{bmatrix}$$

Solve Row 2:

$$-T_1 + 2T_2 - T_3 = -10 \implies -100 + 2T_2 - 100 = -10$$

$$2T_2 - 200 = -10 \implies 2T_2 = 190 \implies \mathbf{T_2 = 95}$$

Physical Intuition Check: Both edges are being held at 100. But because the inside of the bar is actively sucking energy out of the system ($Q = -10$), the middle of the bar dips below the boundary temperatures ($T_2 = 95$). Heat continuously flows inward from both boiling edges.


Example 3: Endothermic Fridge ($Q < 0$, Both Ends at 100°C)
Analytical (exact)
FEM nodes
Boundary temp (100°C)
107.5
105.0
102.5
100.0
97.5
95.0
92.5
90.0
Temperature T (°C)
$T = 100$°C
$T = 95$°C
$T = 100$°C
Node 1
($x = 0$)
Node 2
($x = h$)
Node 3
($x = 2h$)
Position x (m)

Why Are Nodal Errors Zero?

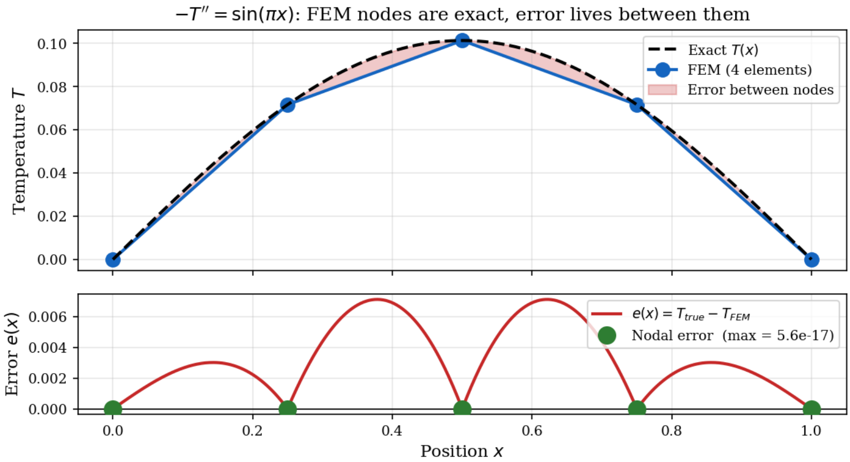


Figure: Note that nodal errors will be zero for arbitrary $Q(x)$, which will be proven soon!

Cont.

It seems kind of weird that the errors at the nodes are exactly zero. This is not general; it's actually due to some interesting properties of the FEM solution, and one method of showing errors to be zero follows.

Let T_{true} be the exact analytical temperature and T_{pred} be our FE approximation. Both must satisfy the weak form for any test function N_i:

$$\int_0^L k \frac{dT_{true}}{dx} \frac{dN_i}{dx} dx = \int_0^L QN_i dx$$

$$\int_0^L k \frac{dT_{pred}}{dx} \frac{dN_i}{dx} dx = \int_0^L QN_i dx$$

Cont.

Because both equal the same heat source integral, we can subtract the predicted equation from the true equation:

$$\int_0^L k \left(\frac{dT_{true}}{dx} - \frac{dT_{pred}}{dx} \right) \frac{dN_i}{dx} dx = 0$$

Defining our discretization error as $e = T_{true} - T_{pred}$, we arrive at the fundamental property of **Galerkin Orthogonality**:

$$\int_0^L k \frac{de}{dx} \frac{dN_i}{dx} dx = 0$$

Evaluating the Nodal Error

Assuming a uniform mesh size h, the derivative of our linear shape function N_i is piecewise constant:

$$\frac{dN_i}{dx} = \begin{cases} 1/h & \text{on } [x_{i-1}, x_i] \\ -1/h & \text{on } [x_i, x_{i+1}] \end{cases}$$

Splitting our orthogonality integral over these two regions:

$$\int_{x_{i-1}}^{x_i} k \frac{de}{dx} \left(\frac{1}{h}\right) dx + \int_{x_i}^{x_{i+1}} k \frac{de}{dx} \left(-\frac{1}{h}\right) dx = 0$$

Cont.

By the Fundamental Theorem of Calculus ($\int e' dx = e$), we evaluate the boundaries exactly at the nodes:

$$\frac{k}{h}\Big[e(x_i) - e(x_{i-1})\Big] - \frac{k}{h}\Big[e(x_{i+1}) - e(x_i)\Big] = 0$$

Dividing out k/h and simplifying the algebra gives:

$$2e(x_i) = e(x_{i-1}) + e(x_{i+1})$$

The Boundary Argument: Why the error vanishes

The relation $2e(x_i) = e(x_{i-1}) + e(x_{i+1})$ means every nodal error is the perfect average of its neighbors.

Geometrically, a discrete sequence of points where every value is the average of its neighbors must form a **straight line**. To define this line, we look at the boundaries:

- **Dirichlet BC (e.g., at x_0):** The FE solution is forced to exactly match the known boundary value. Therefore, the error is anchored at zero: $e(x_0) = 0$.
- **Neumann BC (e.g., at x_n):** Applying orthogonality to the half-shape function at a free boundary yields $e(x_n) - e(x_{n-1}) = 0$. The error line has a slope of zero.

Conclusion: Must be zero at one point, and zero derivative everywhere, hence zero everywhere.

Wow! FEM is so Amazing!

There's a catch. Of course there is. This "perfect error at nodes" is extremely fragile. Imagine the bar starts losing energy to surrounding air:

$$-k\frac{d^2T}{dx^2} + \beta T = Q$$

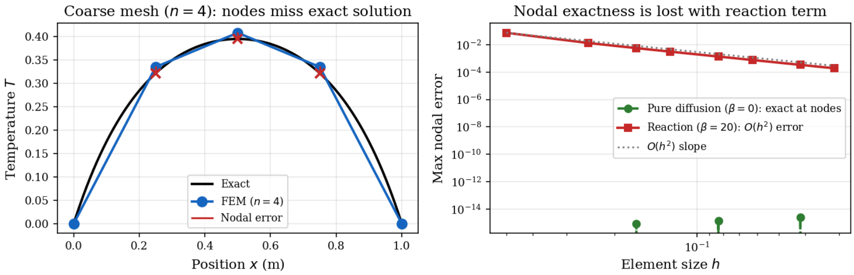
Now there's going to be this term

$$\frac{\beta h}{6}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}\{T\}$$

which predictably breaks the proof earlier.

Cont.

Adding βT term: $-kT'' + \beta T = Q$ breaks the Green\'s function proof ($\beta = 20$)



Appendix: Intuition

What does K_{ij} actually represent?

In engineering, $[K]$ is called the **Stiffness Matrix**. In our heat problem, it represents **Mutual Influence**.

$$K_{ij} = k \int_0^h \frac{dN_j}{dx} \frac{dN_i}{dx} dx$$

- **The Physical Meaning:** K_{ij} tells us how much the temperature at Node j affects the temperature at Node i.
- **Diagonal (K_{ii}):** "Self-influence." How much a node resists temperature change based on its own conductivity.
- **Off-Diagonal (K_{ij}):** "Connection." Since our nodes are connected by a bar, a hot Node j will naturally try to pull Node i's temperature up.

Note: If $K_{ij} = 0$, it means Node i and Node j aren't neighbors. They don't know each other exist! (This is why global matrices are "sparse").

The Load Vector F_i (The Collection Plate)

If $[K]$ is the structure of the bar, $\{F\}$ is what the **outside world** is doing to it.

$$F_i = \int_0^h Q \cdot N_i(x)dx$$

- **The Concept:** A node isn't actually just a point; It represents the point and the immediate area around it. N_i is like a collection plate to catch internal heat Q.
- **The weighting:** Because N_i is strongest at node i and tapers off, node i gets the heat that happens closest to it.
- **The Result:** For a constant Q on a slice of length h, the math gives us $Qh/2$.

Conservation of Energy

If we look at a single row i of our finished system $[K]\{T\} = \{F\}$, we can rearrange the math to see exactly what is happening to Node i:

$$\underbrace{K_{ii}T_i}_{\text{Heat of Node}} = \underbrace{F_i}_{\text{External Heat}} + \underbrace{\sum_{j\neq i}(-K_{ij}T_j)}_{\text{Neighbor Pressure}}$$

- **Internal State ($K_{ii}T_i$):** This is the node's internal heat.
- **External Reality (F_i):** Direct heat being pumped in. (Note that because this is steady-state, ignoring neighbor pressure inner heat will equalize with outer heat.)
- **Neighbor Pressure ($\sum -K_{ij}T_j$):** Because off-diagonal K_{ij} values are negative, this term becomes positive, representing that heat will flow from neighbors into the node.